Matthew Faw

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Research Interests

Stochastic Optimization, Online Decision-Making, Reinforcement Learning and Control

Education

2024–present **ARC Postdoctoral Fellow**, *Georgia Institute of Technology*, Atlanta, GA.
Mentors: Siva Theja Maguluri, Sahil Singla

2018–2024 **Ph.D. ECE (Thesis: Adaptive Algorithms for Stochastic Optimization and Bandit Learning)**, *The University of Texas at Austin*, Austin, TX.
Advisors: Sanjay Shakkottai, Constantine Caramanis.

2013–2017 **B.S.E. Electrical & Computer Engineering, B.S. Computer Science, A.B. Math**, *Duke University*, Durham, NC.
Advisors: Nick Buchler, Richard Fair, Benjamin C. Lee

Publications (Google Scholar)

In Submission

2026 "Online Allocation Using Few Samples", **F**, S. Singla, Y. Wang (alphabetical order)

2026 "Online Control with Multiple Sensors", **F**, S. T. Maguluri

Conference Papers

ICML 2025 "On Mitigating Affinity Bias through Bandits with Evolving Biased Feedback", **F**, C. Caramanis, J. Hoffmann

ICML 2025 "In-Context Fine-Tuning for Time-Series Foundation Models", **F**, R. Sen, Y. Zhou, A. Das

COLT 2023 "Beyond Uniform Smoothness: A Stopped Analysis of Adaptive SGD", **F**$^{=}$, L. Rout$^{=}$, C. Caramanis, S. Shakkottai

COLT 2022 "The Power of Adaptivity in SGD: Self-Tuning Step Sizes with Unbounded Gradients and Affine Variance", **F**$^{=}$, I. Tziotis$^{=}$, C. Caramanis, A. Mokhtari, S. Shakkottai, R. Ward

SIGMETRICS 2022 "Learning To Maximize Welfare with a Reusable Resource", **F**$^{=}$, O. Papadigenopoulos$^{=}$, C. Caramanis, S. Shakkottai

SODA 2022 "Single-Sample Prophet Inequalities via Greedy-Ordered Selection", C. Caramanis, P. Dütting, **F**, P. Lazos, S. Leonardi, O. Papadigenopoulos, E. Pountourakis, R. Reiffenhäuser (alphabetical order)

NeurIPS 2020 "Mix and Match: An Optimistic Tree-Search Approach for Learning Models from Mixture Distributions", **F**, R. Sen, K. Shanmugam, C. Caramanis, S. Shakkottai

Journal Papers

Theory of Computing 2026 "Single-Sample Prophet Inequalities via Greedy-Ordered Selection", C. Caramanis, P. Dütting, **F**, P. Lazos, S. Leonardi, O. Papadigenopoulos, E. Pountourakis, R. Reiffenhäuser (alphabetical order)

POMACS 2022 "Learning To Maximize Welfare with a Reusable Resource", **F**$^{=}$, O. Papadigenopoulos$^{=}$, C. Caramanis, S. Shakkottai

TOCS 2017 "Computational Sprinting: Architecture, Dynamics, and Strategies", S. Zahedi, S. Fan, **F**, E. Cole, B. Lee

Awards + Honors

2023 Dr. Brooks Carlton Fowler Endowed Presidential Graduate Fellowship in ECE, 2023-2024 academic year

2022 Top 10% reviewer for NeurIPS'22 and AISTATS'22, Highlighted reviewer for ICLR 2022

2020 NXP Foundation Fellowship, 2020-2021 academic year

2017 Cum Laude Graduation Honors, Duke University

2016 Member, Tau Beta Pi and Eta Kappa Nu Honor Societies, Duke University

2014 Gold medal, International Genetically Engineered Machine Competition

Talks and Poster Presentations

Invited Talks

October 2025 INFORMS Job Market Showcase, Atlanta, GA: "Optimal Control with Multiple Sensors"

July 2025 INFORMS APS Conference, Atlanta, GA: "Fundamental Limits of Regret Minimization in Stochastic Bandits with Evolving Biased Feedback"

February 2025 Google Research, (Virtual): "Order-Optimal Convergence Rates with Adaptive SGD"

October 2024 INFORMS Annual Meeting, Seattle, WA: "Order-Optimal Convergence Rates with Adaptive SGD"

March 2024 Georgia Tech ARC Colloquium, Atlanta, GA: "The Power of Adaptivity in SGD"

Talks

April 2026 Applied Probability Group Meeting, Atlanta, GA: "Online Control with Multiple Sensors"

July 2023 COLT 2023, Bangalore, India: "Beyond Uniform Smoothness: A Stopped Analysis of Adaptive SGD"

April 2023 IFML Workshop, UW "Beyond Uniform Smoothness: A Stopped Analysis of Adaptive SGD"

July 2022 COLT 2022, London, UK: "The Power of Adaptivity in SGD: Self-Tuning Step Sizes with Unbounded Gradients and Affine Variance"

June 2022 SIGMETRICS 2022, IIT Bombay, Mumbai, IN: "Learning To Maximize Welfare with a Reusable Resource"

April 2022 Machine Learning Lab Research Symposium, UT Austin: "The Power of Adaptivity in SGD: Self-Tuning Step Sizes with Unbounded Gradients and Affine Variance"

January 2022 SODA 2022, Virtual: "Single Sample Prophet Inequalities via Greedy-Ordered Selection"

Poster Presentations

July 2025 ICML 2025, Vancouver, Canada, "In-Context Fine-Tuning for Time-Series Foundation Models" and "On Mitigating Affinity Bias through Bandits with Evolving Biased Feedback"

May 2024 IFML NSF Site Visit Poster Session, UT Austin, "Beyond Uniform Smoothness: A Stopped Analysis of Adaptive SGD"

December 2023 NeurIPS 2023 Workshop: Algorithmic Fairness through the Lens of Time, New Orleans, LA, "On Mitigating Affinity Bias through Bandits with Evolving Biased Feedback"

October 2022 Joint IFML/Data-Driven Decision Processes Workshop, Simons Institute, UC Berkeley, "The Power of Adaptivity in SGD: Self-Tuning Step Sizes with Unbounded Gradients and Affine Variance"

December 2020 NeurIPS 2020, Virtual, "Mix and Match: An Optimistic Tree-Search Approach for Learning Models from Mixture Distributions"

November 2019 Texas Wireless Summit, UT Austin, "Mix and Match: An Optimistic Tree-Search Approach for Learning Models from Mixture Distributions"

Conference Reviewing

2021-Present AISTATS, ALT, ICLR, ICML, JMLR, NeurIPS

Industry Experience

March–Sept. 2025 **Visiting Researcher**, *Google Research*, Remote.

June-Sept. 2024 **Ph.D. Student Researcher**, *Google Research*, Mountain View, CA.

June'17-July'18 **Software Engineer**, *Verato*, McLean, VA.

May-Aug 2016 **Software Engineering Intern, Stateflow Semantics**, *MathWorks*, Natick, MA.

Undergraduate Research Experience

Jan-Dec 2016 **Datacenter Architecture**, *Advisor: Dr. Benjamin Lee*, Duke University.

Jan-Dec 2015 **Microfluidics**, *Advisor: Dr. Richard Fair*, Duke University.

May-Nov 2014 **Synthetic Biology**, *Advisor: Dr. Nick Buchler*, Duke University.

Teaching Experience

Spring 2026 **ISyE 3770 Statistics and Applications Instructor**, *Georgia Tech*.
Served as the instructor for introductory probability and statistics course. Designed and delivered lectures, office hours, homework, and exams for 60 undergraduate students. Managed two undergraduate TAs.

Fall 2018, Spring 2019 **EE 460J, Data Science Lab TA**, *UT Austin*.
Led lab sessions, and graded homeworks and exams

Spring 2017 **CS 308, Software Design and Implementation TA**, *Duke University*.
Personally mentored 3 undergraduate CS students, conducted code reviews, advised and evaluated design and implementation of 3 software projects.

Fall 2015 **ECE 280, Signals & Systems TA**, *Duke University*.
Held office hours and graded assignments for 70 undergraduate students.

Spring 2015 **Synthetic Biology House Course Co-Designer/Instructor**, *Duke University*.
Co-designed and taught the first-offered Duke course on synthetic biology to 10 undergraduates.

Graduate Coursework

UT Austin Probability & Stochastic Processes, Advanced Probability, Stochastic Processes I, Theoretical Statistics, Online Learning, Large Scale Optimization I & II, Combinatorial Optimization, Sublinear Algorithms, Markov Chains & Mixing Times, Combinatorics & Graph Theory, Analysis & Design of Communication Networks

Technical Skills

Programming Python, Java, C/C++, JavaScript

Infrastructure Kubernetes, AWS, Google Cloud, Mongo, Solr

References

Sanjay Shakkottai, UT Austin, sanjay.shakkottai@utexas.edu

Constantine Caramanis, UT Austin, constantine@utexas.edu

Siva Theja Maguluri, Georgia Institute of Technology, siva.theja@gatech.edu